

IMPERIAL



Recent changes in the Firedrake software ecosystem

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Firedrake '22 pyop3: A new domain-specific language for automating high-performance mesh-based simulation codes

Firedrake '23 Latest developments in pyop3

Firedrake '24 pyop3 is coming

Firedrake USA '25 pyop3: a better code generator for mesh-based computations



Submit thesis/Begin postdoc

10/24



New Mac runners

11/24



Absorb PyOP2 and TSFC

12/24



pip install firedrake

02/25



How did we get here?

- Firedrake needs a particular environment of compiled dependencies
- HPC makes this harder
- Hence `firedrake-install`



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- `firedrake-install` was not a robust solution
- Install failures are extremely hard to debug



What happened?

- `pip install libsupermesh`
- 3 stage install with `firedrake-configure` (intentionally stupid)
- OpenMPI instead of MPICH (`firedrake-run-split-tests`)
- `firedrake-check`
- `--netgen` etc into `pyproject.toml`
- New Docker images



Success stories

- Much easier HPC install
- PETSc `--download-firedrake`
- NixOS package



New Linux runners

02/25

2 × ...

- 384GB RAM
- NVIDIA A100 GPU
- 4 runners



Smoke tests

02/25

- Gusto
- Thetis
- G-ADOPT
- Spyro

It really works!



How to add smoke tests for your library

```
- name: Run Gusto smoke tests
  # Only test Gusto in real mode
  if: |
    inputs.run_tests &&
    (success() || steps.install.conclusion == 'success') &&
    matrix.arch == 'default'
  run: |
    . venv/bin/activate
    git clone --depth 1 https://github.com/firedrakeproject/gusto.git gusto-repo
    pip install --verbose ./gusto-repo
    python -m pytest -n 8 --verbose \
      gusto-repo/integration-tests/balance/test_saturated_balance.py \
      gusto-repo/integration-tests/equations/test_thermal_sw.py \
      gusto-repo/integration-tests/transport/test_embedded_dg_advection.py
    timeout-minutes: 10
```



02/25



Stop using our own fork of PETSc

03/25



First Firedrake release

04/25



How it happened

- Coordinated the release of UFL, pyadjoint, loopy
- Made the first releases of FIAT and libsupermesh
- 2025.4.0 released on PyPI



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- Coordinated the release of UFL, pyadjoint, loopy
- Made the first releases of FIAT and libsupermesh
- 2025.4.0 released on PyPI
- It immediately broke
- 2025.4.0.post0 released



Understanding Firedrake's release strategy



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2025.4.0 release

main



Understanding Firedrake's release strategy

	docs	breaking change 1
	bugfix 1	new feature 1
2025.4.0	release	main

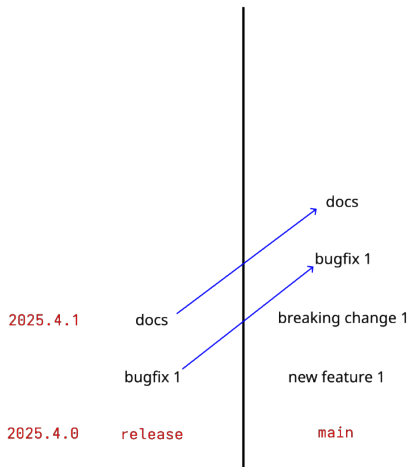


Understanding Firedrake's release strategy

2025.4.1	docs	breaking change 1
	bugfix 1	new feature 1
2025.4.0	release	main



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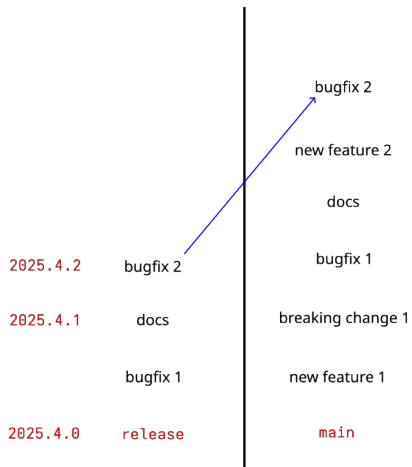
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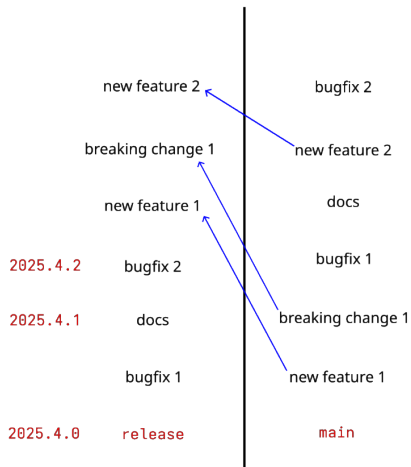
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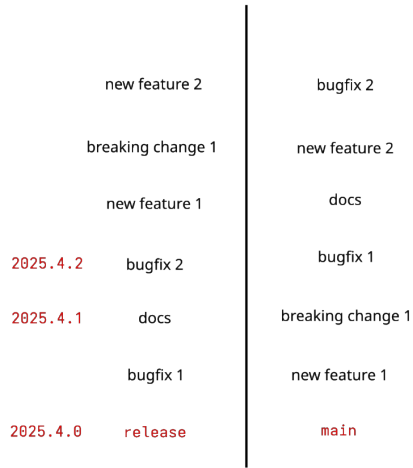
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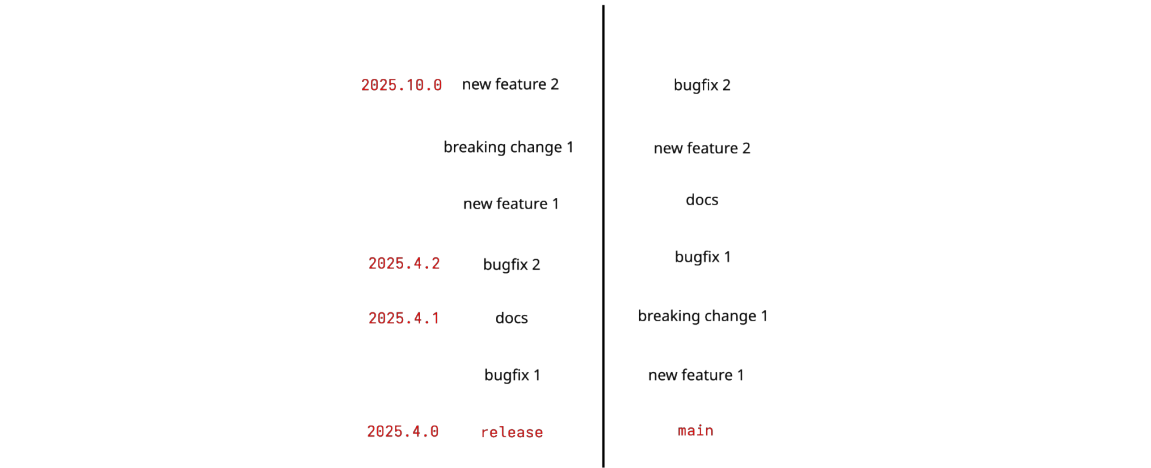
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Supporting main and/or release



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Strategy 3 Check the version.

```
if importlib.metadata.version("firedrake") > "2025.4":  
    # new API  
else:  
    # old API
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We have Docker images for release (firedrake-vanilla-default:dev-release) and main (firedrake-vanilla-default:dev-main) that you can use for your CI.



Use more system packages when building PETSc

04/25



- Interacting with PETSc options: `OptionsManager`
- Passing arbitrary data through PETSc to Python preconditioners: 'app context'
- Handling citations
- Help initialising PETSc
- Reading PETSc configuration information

<https://firedrakeproject.org/petsctools>



Final remarks

- A lot has changed in Firedrake in the past year
- Feedback welcome on what the community wants next (`pip install petsc?`, `conda?`, `spack?`)

